Where the S5 negative space lands

Follow-up: from an 800-query hole to a saturation measurement

The first deck ended with the 800-query whole-skeleton negspace holding at zero across 5.4M structures, and the full 6,336-cell sweeps in flight. They are now complete. We then ran every fold-novelty probe we could reach — AFDB dark matter, TED novel folds, and a multi-filter annotation funnel — through the same S5 instrument. All land on known geometry. One probe looked like it broke the pattern; it turned out to be a methodological trap worth stating carefully, because it is the objection a skeptic will raise.

Recap — where deck 1 left off

- Enumeration: 198 S5 skeletons × 32 H/E typings = 6,336 queries from the Chitturi 2016 2D-hex-lattice model.
- Whole-skeleton negspace: 25 skeletons (800 queries) return zero across every typing.
- Scale-up test: 800 queries × (19k curated + 496k experimental PDB + 4.9M AFDB) = 2,400 sweeps, 0 lit-up. Not one negspace motif is realized at 5.4M structures.
- Cost: four latent searchmatrix buffer/validation bugs, all fixed.
- Then in flight: the full 6,336-cell sweeps + the interactive inspector.

Deck 2 picks up exactly here: the sweeps finished, and the question became how far does the negative generalize?

The full landscape is in — four databases, one matrix

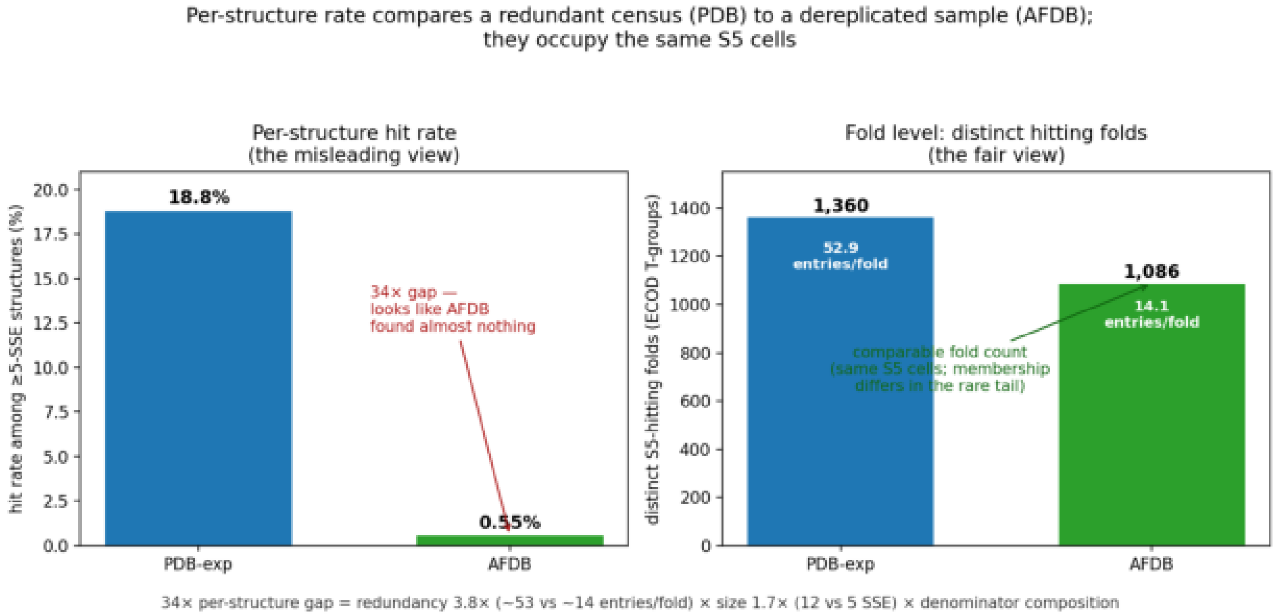
- 3,380 of 6,336 cells stay empty across all four (the per-cell negspace; a superset of the 800 whole-skeleton holes).
- All browsable in the inspector: heatmaps, per-cell exemplars rolled up by ECOD T-group, real per-pair contact geometry, all-pairs compositional enrichment.

The reframe — the negative is the result

- S5 is an assignment-free instrument: pure SSE geometry, fully enumerable, so it comes with a denominator (6,336). That makes "what's missing" a measurable quantity, not an anecdote.
- A novelty test built on CATH/ECOD begs the question (novel = "we didn't classify it"). S5 does not.
- The claim: the observable 5-SSE fold repertoire is effectively closed. AlphaFold multiplies the membership of known folds by orders of magnitude but adds essentially no new local topology.

Sequence space is exploding; it maps many-to-one onto a bounded, nearly-filled fold codomain that stopped growing orders of magnitude of sequence ago. AF is the lens that lets you see the collapse.

AFDB coverage: per-structure misleads, fold level agrees

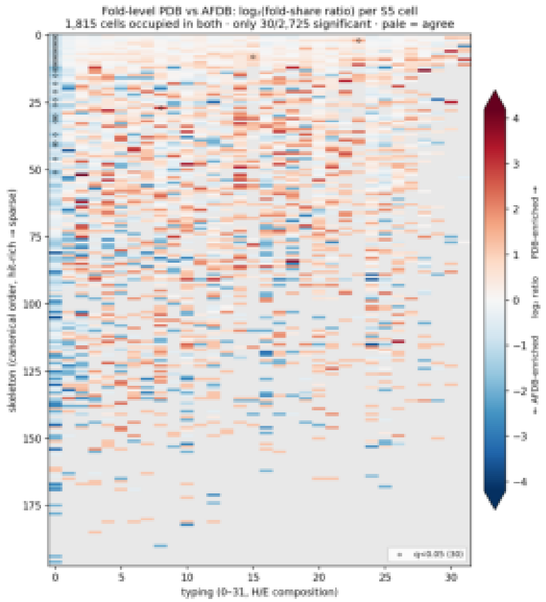


• The trap: per-structure hit rate looks like AFDB found almost nothing — PDB-exp 18.8% vs AFDB 0.55% among ≥5-SSE structures (34×). But that compares a redundant crystallization census to a dereplicated, smaller diversity sample — not a fair comparison.

• The fair view (geometry): PDB and AFDB occupy the same S5 cells (~88% of AFDB's cells shared) and hit a comparable number of folds — PDB 1,360 vs AFDB 1,086 T-groups. (Fold membership overlaps only ~⅓ — a rare-tail sampling difference — but it carries no unique cells either way.)

• The 34× decomposes: redundancy 3.8× (PDB ~53 entries/fold vs AFDB ~14) × size 1.7× (AFDB search set is dereplicated cluster reps, and cluster-space is dominated by small-domain families → median 5 vs 12 SSE) × denominator composition (AFDB "capable" = the whole dereplicated proteome, mostly small/dark non-fold clusters). Not incompleteness — the sweep is complete (6,336 queries, 4.92M entries, verified).

Cell by cell, PDB and AFDB agree at the fold level



• Redundancy-controlled diff: $\log_2$(fold-share PDB / AFDB) per S5 cell, same skeleton×typing grid. Pale = agree.

• 1,815 cells occupied in both; only 30 of 2,725 differentially enriched ($q<0.05$, circled). A faint PDB lean (it carries ~25% more total folds) — but no cell class is AFDB's alone, and none PDB's alone at scale.

• Two independent samples of protein space — a redundant crystallographic census and a dereplicated predicted proteome — occupy the same motif cells in the same proportions. That convergence is the real evidence for a bounded repertoire.

• The AFDB-enriched cells are all HHHHH (all-α) — and it's a confound, not new diversity: the same α-solenoid/repeat families (ARM, Ankyrin, HEAT — X-group 109) + TM-helical (GPCR, MFS) tile overlapping 5-helix windows across many skeletons at once, so one family inflates every all-α cell. Composition (eukaryote repeats/TM) × S5-blind-to-helical-identity × solenoid tiling — not reachable topology.

Why the fold repertoire looks closed — and the honest caveat

- The cell-level agreement (AFDB reaches the same S5 cells as PDB, from a
- AFDB's lower count (1,086 vs 1,360) is shallower sampling of the rarer folds,
- Caveat kept in view: "fold" here = ECOD T-group as the dereplication unit;

totally independent, dereplicated sample) is the positive statement: the two views of protein space converge on the same bounded set of local motifs.

not a different repertoire — it adds essentially no cells PDB lacks.

the per-structure rate is genuinely uninformative across datasets of different redundancy + size, so all cross-dataset claims are made at the fold level with the completeness audit attached.

The novelty gauntlet — every probe we could reach

- Four independent definitions of novelty — geometric, dark-sequence, no-CATH, and annotation-filtered.
- All land on known S5 geometry. The one apparent exception (v3) is a methodological artifact — next slides.

Each pushes a differently-defined "new fold" set through the same S5 instrument.

Probes 1–2 — dark matter and TED novel folds

- ~16% of the DB but only ~4% of hits — it under-hits S5.
- 786 occupied cells, 96% already PDB-known; ~16 novel-vs-all-known, all singletons.
- 89% land on PDB-occupied cells.
- Exactly 1 robust (multi-structure) novel cell: skeleton 118 / EEHHE.
- Non-circular: S5 knows nothing about CATH, yet agrees the geometry is known.

AFDB dark matter (sequence-dark ≠ fold-dark):

TED novel folds (defined by no CATH match — a classification-based novelty claim):

Even the most-curated fold-novelty set on the planet is, at the 5-SSE level, geometry we have already seen.

Probe 3 — the annotation funnel, and the trap

- ~/work/afdb_200M GENUINE funnel: dark clusters → no-Pfam → no-CATH → no-InterPro → pLDDT≥70 → low-decay → not-prophage → ≥10 members. 4,201 "genuine novelty" candidate domains.

- Searched as whole multi-domain models, they appear to hit 40 robust never-seen cells (top: HEEHE, 6 independent proteins) — the only probe to break the pattern.

- The catch: those motifs draw their five SSEs from scattered positions across a whole protein (median span 234 res vs 153 for known-cell hits) — i.e. across domain boundaries. An arrangement a domain-chopped reference search structurally cannot produce.

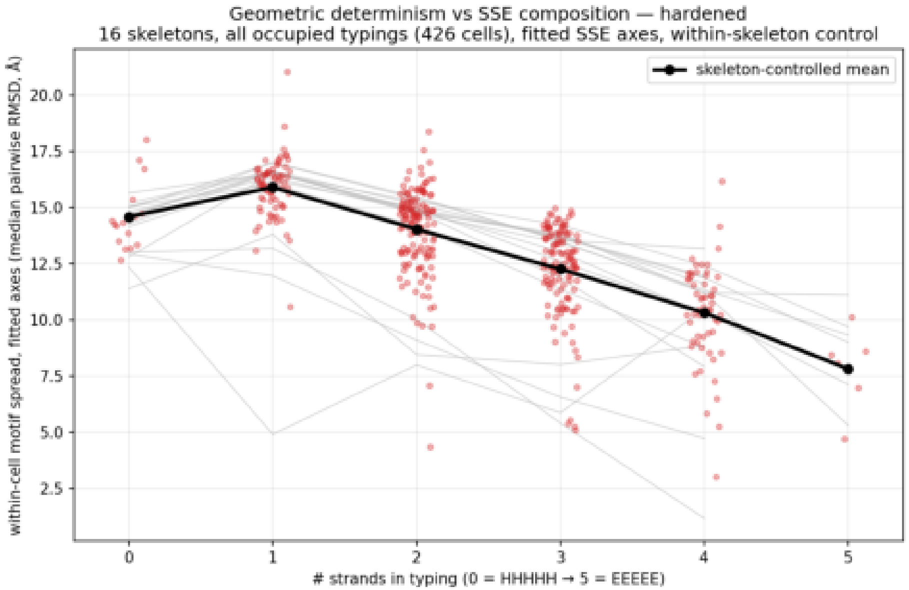
The reference databases were searched as chopped domains; v3 was searched as whole models. That granularity mismatch — not novelty — manufactures never-seen cells.

The sealing test — a controlled A/B

- The 40 robust "novel" cells → 0. The 25 residual never-seen cells are all singletons (noise floor).
- Lesson for the paper: whole-model vs domain-chopped input must be matched, or cross-domain SSE assembly masquerades as fold novelty.

Chop each candidate to its own committed _D<n> domain boundary (the pipeline's own decomposition — no arbitrary cutoff), rebuild with the identical PALSSE + generateMatrix pipeline. Only one variable changes: chopping.

When to trust S5 — the instrument's own limits



• Composition-graded fidelity: within-cell 5-SSE geometry tightens monotonically with strand count (~−1.9 Å/strand); a 1-strand penalty — all-α / single-strand cells are grab-bags, β-registration makes a cell determinate.

• It's baked into the query, not just the structures: an EEEEE query carries ~700 hard - (non-contact) constraints + ~1,000 sheet-pairing codes (c/t) across the 198 skeletons; HHHHH is ~854 X wildcards with zero hard constraints. Same-sheet non-adjacent strands cannot touch (register-separated) → -; non-adjacent helices are unconstrained → X. So all-α cells are intrinsically looser queries and match promiscuously — the fidelity gradient is half structural, half encoding.

• Blind to transmembrane: GPCR (5001.1.1) and MFS (5050.1.1) — maximally different TM folds — both land in generic HHHHH cells.

• Tiling inflates abundance: ARM repeats (109.4.1) — one domain hits up to 123 cells. Abundant ≠ novel:

Where it lands

• "ECOD is completed" — bounded to three regimes: (1) the common core is discovery-complete; (2) the frontier moved from discovery to membership (AF = remote-homology-at-scale); (3) the tail is curation-hard, not discovery-rich (repeats, symmetry, ARM).

• Bottom line: AFDB does not robustly reach S5 cells PDB can't — its cells are an ~88% subset of PDB's (a geometry subset, not a fold subset), sampled far more densely, and every novelty probe lands on known geometry. Even PDB's unique folds add no new cells (e.g. Ig sub-family 11.9.1: 46/46 cells AFDB-occupied). Two caveats bound this:

• Caveat 1 — S5 is a local, fixed-cardinality descriptor (Lesk): it measures 5-SSE local motif geometry — blind to global topology, transmembrane arrangement, domain-scale novelty. "No new S5 cells" ≠ "no novel global folds"; an empty cell = "never observed here," not "unfoldable."

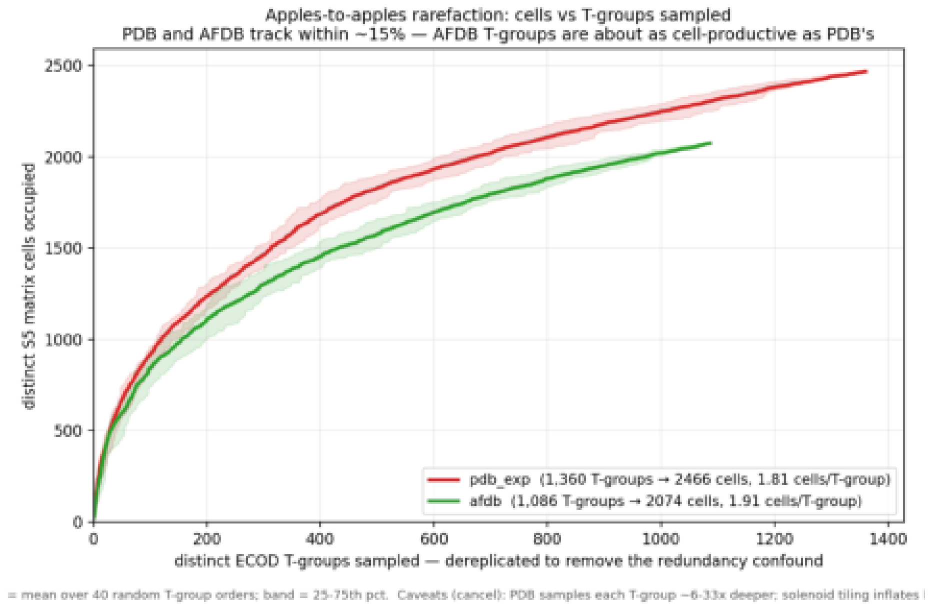
• Caveat 2 — AlphaFold was trained on the PDB: AFDB agreeing with PDB on what's reachable is partly a training prior. The data can't cleanly separate "the real proteome is saturated" from "AF only renders PDB-like geometry." Same practical result (broader prediction surfaces no new cells), genuinely ambiguous mechanism.

• Still open: are the 3,380 empty cells sterically reachable-but-untaken, or lattice over-generation? Resolution needs constructive 3D backbone modeling (RFdiffusion / Rosetta) on the negspace.

Recap

- Full landscape done: 6,336 cells × 4 databases; 3,380 stay empty; all in the interactive inspector.
- The negative reframed: a saturation measurement with an assignment-free, enumerable instrument — the observable 5-SSE repertoire is effectively closed.
- PDB ≈ AFDB in geometry: per-structure hit rate misleads (34×, redundancy×size); PDB and AFDB occupy the same S5 cells and hit comparable fold counts (1,360 vs 1,086) — T-group membership diverges only in the rare tail (no new cells). Coverage is redundancy, not incompleteness.
- Novelty gauntlet: four independent novelty definitions (lattice, dark-sequence, no-CATH, annotation-funnel) → all land on known geometry.
- The one trap: whole-model search fabricated 40 "novel" cells; a controlled domain-chopped re-search collapsed them to 0. Input granularity, not topology.
- Deliverables: inspector (leda:3001/docs), sealed capstone, full session log.
- Open: geometric reachability of the 3,380 — constructive modeling, not more search.

Backup — apples-to-apples rarefaction (per T-group)



• The honest replacement for the retired per-structure rarefaction: x-axis is distinct ECOD T-groups, not structures → removes the redundancy confound (PDB census vs AFDB dereplicated).

• PDB and AFDB track within ~15% — AFDB T-groups are about as cell-productive as PDB's (1.91 vs 1.81 cells/T-group). No dataset samples matrix-space dramatically better per T-group.

• Band = IQR over 40 random T-group orderings (tight → order-robust). Residual caveats partially cancel: PDB samples each T-group ~6–33× deeper (union → more cells); solenoid tiling inflates cells/T-group in both.